This ICCV paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

CoSMIC: Continual Self-supervised Learning for Multi-Domain Medical Imaging via Conditional Mutual Information Maximization

Yihang Liu[1] Ying Wen[2] Longzhen Yang[1] Lianghua He[1,*] Heng Tao Shen[1]
[1]Tongji University, [2]East China Normal University, * Corresponding author
{2111131, yanglongzhen, helianghua}@tongji.edu.cn, ywen@cs.ecnu.edu.cn, shenhengtao@hotmail.com

Abstract

*Medical foundation models, pre-trained on diverse data sources, have shown significant potential for multi-domain medical imaging tasks. However, the domain shifts across different anatomical types significantly hinder their performance compared to domain-specific models. To address this challenge, we propose CoSMIC, a **Co**ntinual **S**elf-supervised learning framework for **M**ulti-domain med**I**cal image analysis, with the core idea of **C**onditional mutual information maximization. Specifically, CoSMIC (i) acquires domain-specific knowledge sequentially, bypassing domain shifts caused by joint pre-training; (ii) enhances generalized representations by proposing a novel conditional contrastive loss to prevent catastrophic forgetting. This loss hierarchically aligns multi-view features within the current domain, maximizing their mutual information conditioned on domain-invariant representations extracted from prior domains through Anatomy-Guided Calibration. We pre-train CoSMIC across four medical domains and evaluate it on fifteen downstream datasets from five domains: Retinoscopy, Radiography, Ophthalmoscopy, Dermoscopy, and Histopathology (unseen). Experimental results show that CoSMIC (i) achieves robust feature extraction ability comparable to domain-specific models, (ii) exhibits exceptional generalization capability, significantly surpassing SOTA medical foundation models, and (iii) demonstrates superior transferability to new domains, overcoming current continual pre-training methods.*

1. Introduction

Multi-domain medical imaging aims to develop a unified model capable of generalizing across diverse anatomical domains, such as the chest, eyes, and skin. Recent advances have shown that Self-Supervised Learning (SSL) is a powerful pre-training approach for learning generalized representations that can facilitate a wide range of downstream tasks [5, 25, 45, 57, 70, 71]. However, traditional SSL methods in medical imaging are typically pre-trained on a single anatomical domain, such as fundus images [48, 56, 71] or chest X-rays [19, 38, 57, 68, 69], as illustrated in Fig. 1(a). This domain-specific focus limits their generalizability to downstream tasks across other anatomical regions.

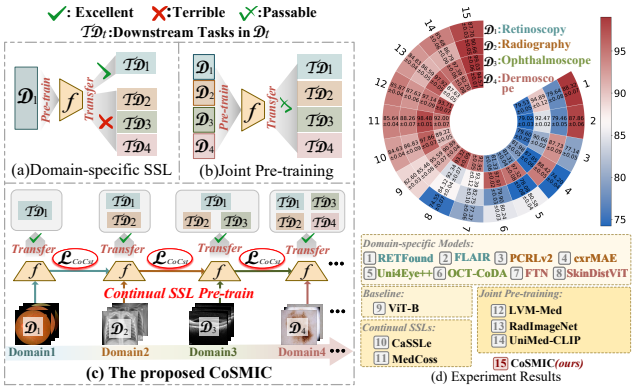


Figure 1. **Comparison of various methods in multi-domain medical imaging tasks**: (a) Models that narrow focus on specific domains; (b) Models that suffer from domain shift, limiting their overall performance on specific tasks; (c) The proposed CoSMIC that enhance both domain-specific and generalized representations for multi-domain medical imaging; (e) Averaged results are presented over 14 downstream datasets from 4 domains.

Medical foundation models, on the other hand, leverage large-scale, multi-modal datasets for joint pre-training, demonstrating a promising advancement in learning generalized representations that can achieve multi-domain medical imaging [35, 40, 41, 58]. However, the inherent heterogeneity of medical images—ranging from differences in imaging modalities (e.g., Microscopy, Tomography, X-rays) to variations in anatomical regions (e.g., skin, eye, chest)—introduces significant domain shifts [63, 72]. These shifts pose a challenge for a single model to generalize effectively across all domains without compromising performance compared to domain-specific models (see Fig.1 (b)).

One promising solution to this issue is Continual Learning (CL) [63], a strategy that enables models to sequentially acquire domain-specific knowledge, thereby adapting to the specific features of each anatomical domain. However, their

effectiveness is restricted by catastrophic forgetting, where learning new patterns generally causes the model to overwrite previously acquired knowledge [17, 49, 60]. Moreover, most existing CLs are task-oriented [6, 15, 17, 28], meaning that the parameters learned at each stage are specific to a particular task and lack transferability to arbitrary tasks within the same domain. Even though a pioneer art has explored combining CL with SSL for multi-domain medical imaging, this approach is heavily reliant on rehearsal-based strategy [63]. Such reliance hinders the model's effectiveness when prior domain data is unavailable, especially in medical data acquisition, which faces privacy and ethical challenges.

To address these issues, we propose CoSMIC, a medical foundation model that unifies **Co**ntinual learning and **S**elf-supervised learning to improve both domain-specific and generalized representations for **M**ulti-domain med**I**cal imaging through **C**onditional mutual information maximization. As illustrated in Fig. 1 (c), CoSMIC is sequentially pre-trained on each anatomical domain to learn specific knowledge, circumventing domain shifts that arise during joint pre-training on diverse datasets. To enhance generalization across domains, CoSMIC introduces a novel conditional contrastive loss ($\mathcal{L}_{CoCst}$) that effectively distills previously learned semantic concepts into current representations, thereby mitigating catastrophic forgetting without relying on historical data replay.

Specifically, instead of traditional distillation-based CL, which directly aligns current representations with their past states [14], $\mathcal{L}_{CoCst}$ in CoSMIC innovatively treats historical knowledge as a conditional variable, thereby constraining the process of maximizing mutual information across multi-view images in the current domain. In practice, within $\mathcal{L}_{CoCst}$: (i) We propose Anatomy-guided Calibration (AGC) to refine token-wise historical representations based on their current embeddings, extracting invariant patterns across model states as conditional variables for robust joint probability prediction. (ii) We treat this joint probability as the clustering center, similarizing intra-image features while discriminating inter-image ones, extending bivariate mutual information to a trivariate space, including past states, thereby effectively distill the old knowledge to current representations. (iii) We introduce hierarchical alignment, computing $\mathcal{L}_{CoCst}$ across network layers to retain multi-scale historical knowledge, enhancing the model's memory capacity and representational consistency.

To summarize, our main contributions are as follows:

- We propose CoSMIC, a medical foundation model that adopts a continual self-supervised learning paradigm to learn both domain-specific and generalized representations for multi-domain medical image analysis.
- In CoSMIC, we introduce conditional contrastive learning, which effectively mitigates catastrophic forgetting within a self-distillation mechanism by maximizing the conditional mutual information of current state representations under the constraint of historical knowledge.
- We pre-train our CoSMIC model across four medical domains—Retinoscopy, Radiography, Ophthalmoscopy, and Dermoscopy—and evaluate its performance on fifteen downstream datasets, including one from the unseen Histopathology domain. Extensive experiments demonstrate CoSMIC's strong scalability, achieving significant improvements across various downstream datasets and outperforming state-of-the-art (SOTA) models, including domain-specific SSLs and unified models with joint pre-training and continual SSLs (Fig. 1 (d)).

2. Related Work

Self-supervised Learning. SSL allows models to learn from data without requiring expert annotations. [2, 21, 22]. In the medical field, SSL is widely used to pre-train models on single-domain data like chest X-rays [20, 51, 57, 68, 69, 74], retinal scans [48, 56, 59, 71], and Optical Coherence Tomography (OCT) [1, 45, 53]. These models can be fine-tuned for domain-specific tasks but struggle to generalize across domains due to narrow focus. For instance, DiRA [20] and Adam [51] require distinct pre-trained weights for retinal and X-ray images. RETFound [71] must be pre-trained separately for retinal scans and OCT, despite both targeting the same anatomical type (eye).

Continual Learning. CL enables the model to acquire new tasks sequentially while preserving previously learned knowledge. This approach encompasses techniques such as rehearsal [3, 6, 61, 63], structure expansion [64, 67], and knowledge distillation [14, 17, 65, 66]. While rehearsal and structure-based methods have shown strong performance, they face challenges in medical image analysis due to the difficulty of acquiring representative data from previous domains [66] and the increased complexity of the model. These limitations make them harder to serve as a versatile tool across highly diverse medical anatomical types.

Medical Foundation Models. The rise of medical foundation models has led to success in multi-domain medical image analysis [36, 40, 41, 62]. These models, trained on large-scale clinical datasets, aim to develop generalized representations through a unified pre-trained network. LVM-Med [41] employs an SSL approach, pre-trained on over 1 million images from 55 medical datasets. RadImageNet [40] is a supervised model trained on 1.3 million medical images. UniMed-CLIP [35] is a vision-language SSL model utilizing 5.3 million image-text pairs. Unimiss+ [58] enhances CT scan analysis by augmenting unpaired Chest X-rays for cross-domain joint SSL training. MedCoss [63] performs a rehearsal-based sequential SSL to learn universal representations across modalities.

Despite advancements in existing work, challenges like

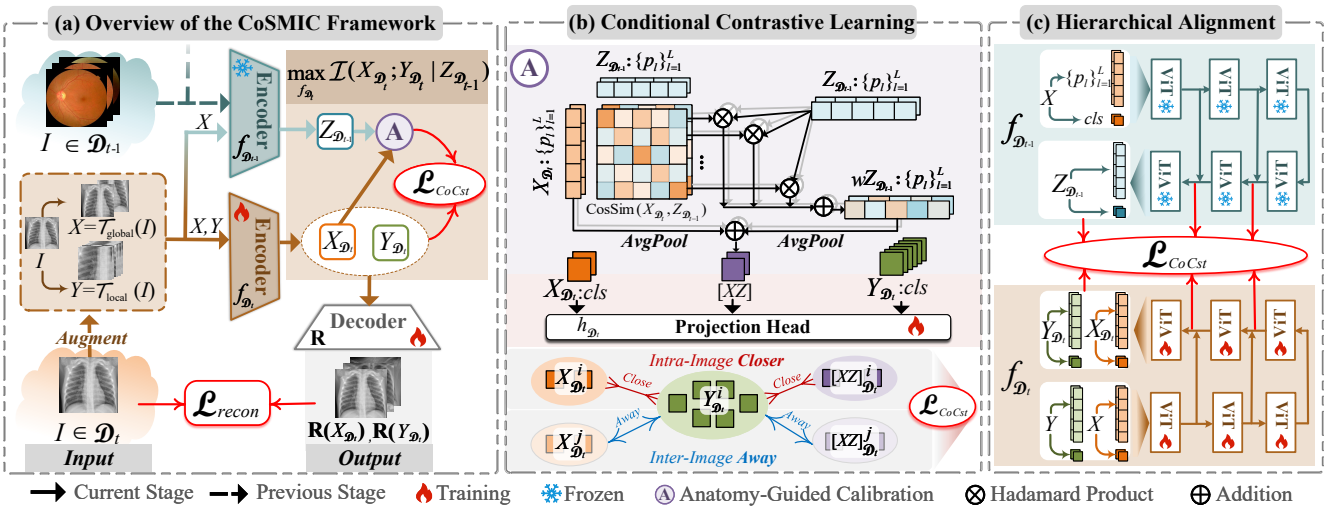


Figure 2. **The illustration of CoSMIC.** (a) In the t-th pre-training stage, CoSMIC acquires domain-specific knowledge by reconstructing latent representations, $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$, to their original form, I, while preserving previously learned patterns by maximizing mutual information between $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$ conditioned on $Z_{\mathcal{D}_{t-1}}$. (b) CoSMIC introduces a conditional contrastive loss, $\mathcal{L}_{CoCst}$, to close the similarity of intra-image pairs, $([XZ]^i_{\mathcal{D}_t}, Y^i_{\mathcal{D}_t})$ and $(X^i_{\mathcal{D}_t}, Y^i_{\mathcal{D}_t})$, while taking away the inter-image pairs, $([XZ]^j_{\mathcal{D}_t}, Y^i_{\mathcal{D}_t})$ and $(X^j_{\mathcal{D}_t}, Y^i_{\mathcal{D}_t})$, based on Eq. 5. (c) $\mathcal{L}_{CoCst}$ is hierarchical aligned across different ViT layers to preserve multi-scale historical patterns.

generalization, domain shift and difficult data acquisition impede their effectiveness in multi-domain medical imaging. Our approach combines the strengths of CL and SSL, introducing a new contrastive loss to maximize conditional mutual information based on historical knowledge, thereby enabling a robust medical foundation model to enhance both domain-specific and generalized representations.

3. Method

The proposed CoSMIC follows the *pre-training* $\rightarrow$ *fine-tuning* protocol, similar to [2, 5, 22, 63]. In the pre-training phase, CoSMIC employs a dual-stream architecture to: (i) continually learn domain-specific knowledge via a reconstruction-based SSL pretext task (Sec. 3.2), and (ii) distill historical knowledge into current representations to enhance generalization through conditional mutual information maximization (Sec. 3.3). During fine-tuning, only the pre-trained feature encoder is used for downstream multi-domain medical imaging tasks.

The overall architecture of CoSMIC is shown in Fig. 2. In the following, we detail each component.

3.1. Preliminary

Given a set of T medical domains $\{\mathcal{D}_t\}_{t=1}^T$, CoSMIC is sequentially pre-trained in each domain and leverages a dual-stream Vision Transformer (ViT) [12] to acquire new knowledge while retaining previously learned patterns. During this process, CoSMIC operates in a self-supervised manner without requiring any data annotation. Specifically, at the t-th stage, as illustrated in Fig. 2(a), when an input image $I \in \mathcal{D}_t$ is received, we first transform it with both global and local image augmentations to generate multi-view images:

$$X = \mathcal{T}_{\text{global}}(I);\ Y = \mathcal{T}_{\text{local}}(I). \tag{1}$$

Empirically, $X = \{x_1, x_2\}$ contains 2 global views and $Y = \{y_1, \ldots, y_6\}$ consists of 6 local views [2].

Both X and Y are then passed through the currently optimized network $f_{\mathcal{D}_t}$, while only the global views from X are passed through the past-learned network $f_{\mathcal{D}_{t-1}}$ (frozen) to extract their token-wise embeddings in both the current state ($X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$) and the past state ($Z_{\mathcal{D}_{t-1}}$). Especially, $\forall x \in X_{\mathcal{D}_t}$, $\forall y \in Y_{\mathcal{D}_t}$, and $\forall z \in Z_{\mathcal{D}_{t-1}}$ take the form of $[cls, \{p_l\}_{l=1}^L]$, where cls is the class token and $\{p_l\}_{l=1}^L$ are image tokens, as defined in [12].

Our goal is to learn a unified pre-trained model ($f_{\mathcal{D}_t}$) that enhances domain-specific representation while retaining the old knowledge in $Z_{\mathcal{D}_{t-1}}$ within $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$.

3.2. Acquiring Domain-specific Knowledge.

CoSMIC implements a simple yet effective reconstruction-based SSL pretext task to acquire domain-specific knowledge at each stage of continual pre-training, inspired by [22, 57, 70, 71, 74]. Specifically, CoSMIC employs a 3-layer ViT as the decoder $\mathbf{R}$ to reconstruct the latent representations $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$ of the corrupted inputs X and Y back to their original form $I \in \mathcal{D}_t$. We use the MSE loss to minimize the reconstruction error between the predicted and original images at the pixel level:

$$\mathcal{L}_{Recon} = \frac{1}{M}(\|\mathbf{R}(X_{\mathcal{D}_t}) - I\|_2^2 + \|\mathbf{R}(Y_{\mathcal{D}_t}) - I\|_2^2), \tag{2}$$

where $M = 8$ refers to the total views of I.

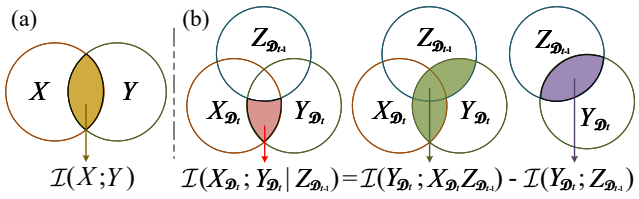


Figure 3. (a) Traditional mutual information. (b) Conditional Mutual Information within trivariate space.

3.3. Enhancing Generalized Representation.

To enable multi-domain medical imaging, CoSMIC employs a self-distillation strategy to mitigate catastrophic forgetting during continual pre-training. In particular, instead of directly making historic representations as the pseudo-label [3, 14, 17], we innovatively propose treating $Z_{\mathcal{D}_{t-1}}$ as the conditional variable to constrain the maximization of mutual information between $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$. This strategy encourages the model to adapt to new domains while reducing error accumulation from noisy pseudo-labels of past domains, thereby enhancing its generalization capability across diverse medical domains.

Specifically, we first provide a mathematical analysis of Conditional Mutual Information Maximization (Sec. 3.3.1). Building on this, we propose a new loss for Conditional Contrastive Learning (Sec. 3.3.2). Finally, we demonstrate the implementation of this loss in CoSMIC (Sec. 3.3.3).

3.3.1. Conditional Mutual Information Maximization

Traditional contrastive learning in SSL maximizes mutual information $\mathcal{I}(X;Y)$ between multiple views X and Y of an input image I (see Fig.3(a)), allowing learning of transferable representations that improve downstream tasks [21, 68]. In CoSMIC, we improve this approach by introducing an additional constraint to preserve previously acquired knowledge. This results in an extension of $\mathcal{I}(X;Y)$ to three probability spaces: $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$, which are extracted by $f_{\mathcal{D}_t}$; and $Z_{\mathcal{D}_{t-1}}$, which is extracted by the frozen $f_{\mathcal{D}_{t-1}}$. We define $Z_{\mathcal{D}_{t-1}}$, which represents the knowledge learned from previous $(t-1)$ domains, as the known condition to maximize mutual information between $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$ in the t-th domain. Consequently, our optimization objective in the t-th domain is to maximize the conditional mutual information:

$$\max_{f_{\mathcal{D}_t}} \mathcal{I}(X_{\mathcal{D}_t}; Y_{\mathcal{D}_t} \mid Z_{\mathcal{D}_{t-1}}). \tag{3}$$

However, direct optimization of Eq.3 is challenging. As an alternative, we decouple it as shown in Fig. 3(b), as follows:

$$\begin{aligned}\mathcal{I}(X_{\mathcal{D}_t}; Y_{\mathcal{D}_t} | Z_{\mathcal{D}_{t-1}}) &= \mathcal{I}(Y_{\mathcal{D}_t}; X_{\mathcal{D}_t} Z_{\mathcal{D}_{t-1}}) - \mathcal{I}(Y_{\mathcal{D}_t}; Z_{\mathcal{D}_{t-1}}) \\ &= \sum_{X_{\mathcal{D}_t}} \sum_{Y_{\mathcal{D}_t}} \sum_{Z_{\mathcal{D}_{t-1}}} P(xyz) \log \frac{P(y|xz)}{P(y|z)}.\end{aligned} \tag{4}$$

Detailed proof is shown in Suppl. 7. Since $\log x$ is a strictly convex function, applying Jensen's inequality to Eq.4 gives:

$$\mathcal{I}(X_{\mathcal{D}_t}; Y_{\mathcal{D}_t} | Z_{\mathcal{D}_{t-1}}) \geq \log \sum_{X_{\mathcal{D}_t}} \sum_{Y_{\mathcal{D}_t}} \sum_{Z_{\mathcal{D}_{t-1}}} P(xyz) \frac{P(y|xz)}{P(y|z)}. \tag{5}$$

Therefore, we only need to optimize this lower bound of $\mathcal{I}(X_{\mathcal{D}_t}; Y_{\mathcal{D}_t} | Z_{\mathcal{D}_{t-1}})$. However, as illustrated in Fig. 3(b), we encounter two key scenarios: (i) when y and the joint representation $[xz]$ are incompatible, Eq. 5 is collapsed as:

$$\log \sum_{Y_{\mathcal{D}_t}} P(y) \sum_{X_{\mathcal{D}_t}} \sum_{Z_{\mathcal{D}_{t-1}}} P(xz) = \log 1 = 0, \tag{6}$$

(ii) when y and x are incompatible, Eq. 5 is collapsed as:

$$\log \sum_{X_{\mathcal{D}_t}} \sum_{Y_{\mathcal{D}_t}} \sum_{Z_{\mathcal{D}_{t-1}}} P(y|z) P(x|z) P(z) = \log 1 = 0. \tag{7}$$

These suggests that the optimization of Eq. 5 is effectively proportional to maximizing the similarity of both $\text{sim}([XZ]_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$ and $\text{sim}(X_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$ for all image views.

3.3.2. Conditional Contrastive Learning

Based on the analysis above, we propose conditional contrastive learning to optimize Eq.3 (See Fig. 2(b)). Specifically, we first design *Anatomy-Guided Calibration* to predict the joint representation $[XZ]_{\mathcal{D}_t}$, then introduce the *Conditional Contrastive Loss* based on InfoNCE loss [43] to similarize pairs of $(Y_{\mathcal{D}_t}, [XZ]_{\mathcal{D}_t})$ and $(X_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$.

Anatomy-Guided Calibration. Noise signals inevitably emerge in $Z_{\mathcal{D}_{t-1}}$ due to distribution shifts between domains [3, 60]. This results in significant error accumulation when $Z_{\mathcal{D}_{t-1}}$ is directly used as pseudo-labels for predicting the joint representation $[XZ]_{\mathcal{D}_t}$. To mitigate this noise, we propose AGC, which predicts $[XZ]_{\mathcal{D}_t}$ by capturing invariant patterns of previously learned semantic concepts that remain robust to changes in the current model's parameters.

Specifically, as shown in Fig.2(b), we utilize the current state information of the same image to calibrate its historical noisy features. For medical images X, despite being processed by different state networks ($f_{\mathcal{D}_t}$ and $f_{\mathcal{D}_{t-1}}$), their anatomical semantic information remains highly consistent. Namely, the anatomical structures corresponding to $p_{l^x} \in X_{\mathcal{D}_t}$ and $p_{l^z} \in Z_{\mathcal{D}_{t-1}}$ exhibit a spatial-semantic correspondence. Here, l^x and l^z are the image token indices of $X_{\mathcal{D}_t}$ and $Z_{\mathcal{D}_{t-1}}$, respectively. This anatomical consistency guides the token-wise alignment of $X_{\mathcal{D}_t}$ and $Z_{\mathcal{D}_{t-1}}$. By computing the cosine similarity between $\{p_{l^x}\}_{l^x}^L$ and $\{p_{l^z}\}_{l^z}^L$, we establish a cross-state weight map as follows:

$$w = \{\{s_t^{l^x,l^z}\}_{l^z}^L\}_{l^x}^L, \quad s_t^{l^x,l^z} = \text{CosSim}(p_{l^x}, p_{l^z}). \tag{8}$$

Then $Z_{\mathcal{D}_{t-1}}$ is calibrated by w to emphasize historical patterns that remain invariant to the state of knowledge in $X_{\mathcal{D}_t}$

Domain	Pre-train		Downstream					
	Dataset	Images	Dataset	Task	#Train	#Val	#Test	Total
$\mathcal{D}_1$:Retinoscopy	EYEPACS	88,702	APTOS	Cls.(5 Classes)	2,048	514	1,100	3,662
			Glaucoma	Cls.(3 Classes)	861	218	465	1,544
			IDRiD	Cls.(5 Classes)	329	84	103	516
	AIROGS	101,442	PAPILA	Cls.(3 Classes)	311	79	98	488
			Retina	Cls.(4 Classes)	336	84	181	601
$\mathcal{D}_2$:Radiography	MIMIC	356,309	NIH	Cls.(14-Multi-label)	78,484	11,212	22,424	112,120
			CXP	Cls.(5-Multi-label)	219,082	5,000	234	224,316
			ZhangCXR	Cls.(2 Classes)	4,983	250	624	5,857
			RSNA	Cls.(2 Classes)	25,184	1,500	3,000	29,684
			SIIM-ARC	Segmentation	7,969	1,898	1,904	11,771
$\mathcal{D}_3$:Ophthalmoscope	OCT2017	84,484	OCTID	Cls.(5 classes)	316	82	174	572
$\mathcal{D}_4$:Dermoscope	ISIC2019	33,769	ISIC2016	Cls.(2 Classes)	750	150	379	1,279
			ISIC2017	Cls.(2 Classes)	2,000	150	600	2,750
	ISIC2020	44,108	ISIC2018	Cls.(7 Classes)	10,015	193	1,512	11,720
			ISIC2018	Segmentation	1,796	300	500	2,596
$\mathcal{D}_5$:Histopathology	-	-	NCH	Cls. (9 classes)	79,994	20,006	7,180	17,180

Table 1. **Summary of the datasets used in CoSMIC.**

when $s_t^{l^x,l^z}$ is higher, and down-weighting potentially noisy patterns when $s_t^{l^x,l^z}$ is lower. This process in the t-th stage can be denoted as:

$$wZ_{\mathcal{D}_{t-1}} = \sum_{l^x=1}^{L} \{s_t^{l^x,l^z} \cdot p_{l^z}\}_{l^z}^{L}. \tag{9}$$

We then predict the joint representation by adding the image tokens of $X_{\mathcal{D}_t}$ and $wZ_{\mathcal{D}_{t-1}}$ after applying AvgPooling:

$$[XZ]_{\mathcal{D}_t} = \text{AvgPool}(X_{\mathcal{D}_t}) + \text{AvgPool}(wZ_{\mathcal{D}_{t-1}}) \tag{10}$$

Conditional Contrastive Loss. We apply a projection head $h_{\mathcal{D}_t}$ to map the cls token of $X_{\mathcal{D}_t}$ and $Y_{\mathcal{D}_t}$, as well as the joint representation $[XZ]_{\mathcal{D}_t}$, into K-dimensional probability distribution space, inspired by [2]. We then employ the InfoNCE loss to minimize the distance between intra-image views and discriminate the distribution of inter-image views of $([XZ]_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$ and $(X_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$.

Specifically, for extracted features a and b from different views of the i-th image, the traditional InfoNCE loss [43] in contrastive learning is defined as:

$$\mathcal{L}_{Cst}(a^i, b^i) = -\log\left(\frac{\delta(a^i, b^i)}{\sum_{j=1}^{B} \delta(a^i, b^j)}\right), \tag{11}$$

where $\delta(a,b) = \exp\left(\frac{\text{sim}(h_{\mathcal{D}_t}(a), h_{\mathcal{D}_t}(b))}{\tau}\right)$, B is the batch size and τ is the temperature hyper-parameter. Then we extend Eq.11 within trivariate space for the i-th image and define the proposed conditional contrastive loss as follows:

$$\begin{aligned}&\mathcal{L}_{CoCst}([XZ]^i_{\mathcal{D}_t}, X^i_{\mathcal{D}_t}, Y^i_{\mathcal{D}_t}) \\ &= \sum_{[XZ]^i_{\mathcal{D}_t}} \sum_{Y^i_{\mathcal{D}_t}} \mathcal{L}_{Cst}([xz]^i, y^i) + \sum_{X^i_{\mathcal{D}_t}} \sum_{Y^i_{\mathcal{D}_t}} \mathcal{L}_{Cst}(x^i, y^i).\end{aligned} \tag{12}$$

3.3.3. Hierarchical Alignment

To preserve multi-scale historical patterns, CoSMIC hierarchically aligns features across ViT layers to compute $\mathcal{L}_{CoCst}$, as illustrated in Fig.2 (c).

Specifically, we use ViT-B [12] as the backbone and introduce skip connections to fuse shallow and deep features for multi-scale feature extraction. In practice, at the $(n+1)$-th layer, the input feature is the sum of the outputs from the n-th and $(N-n)$-th layers, where $n \in (\frac{N}{2}, N]$, and $N = 12$ represents the total number of layers in ViT-B. This process extracts $\frac{N}{2}$ multi-scale features from different ViT layers for each $x \in X_{\mathcal{D}_t}$, $y \in Y_{\mathcal{D}_t}$, and $z \in Z_{\mathcal{D}_{t-1}}$. All of these multi-scale features are used to calculate $\mathcal{L}_{CoCst}$ through alignment in their respective layers. Notably, due to the high computational complexity of $h_{\mathcal{D}_t}$ (with $K = 65,536$ empirically), only features extracted from the N-th layer are passed through $h_{\mathcal{D}_t}$ and compute $\text{sim}(a,b)$ in Eq.12 as cross-entropy. In contrast, features from intermediate layers, where $n \in (\frac{N}{2}, N-1]$, are directly used to compute $\mathcal{L}_{CoCst}$, with $\text{sim}(a,b)$ using MSE similarity instead.

In summary, CoSMIC learns domain-specific knowledge from each stage of continual pre-training through $\mathcal{L}_{Recon}$, while preserving multi-scale historical patterns in the current representations via $\mathcal{L}_{CoCst}$. The overall pre-training objective at the t-th SSL pre-training stage is:

$$\mathcal{L}_{CoSMIC} = \mathcal{L}_{Recon} + \mathcal{L}_{CoCst} \tag{13}$$

4. Experiments

4.1. Implement Ditails

4.1.1. Datasets

We pre-train CoSMIC across 4 medical domains and evaluate its performance in 5 domains, including an unseen domain. Specifically, we define {$\mathcal{D}_1$:Retinoscopy, $\mathcal{D}_2$:Radiography, $\mathcal{D}_3$:Ophthalmoscope, $\mathcal{D}_4$:Dermoscope, $\mathcal{D}_5$:Histopathology (unseen)}. The datasets used in CoSMIC are summarized in Table. 1. In detail, (i) for **continual SSL pre-training**, we utilize EYEPACS [29] and AIROGS [10] for $\mathcal{D}_1$; MIMIC [27] for $\mathcal{D}_2$; OCT2017 [34] for $\mathcal{D}_3$; ISIC2019 [24] and ISIC2020 [47] for $\mathcal{D}_4$. All of these datasets are unlabeled during the SSL pre-training. (ii) For **downstream tasks**, we evaluate CoSMIC on 15 datasets, including APTOS [32], Glaucoma [36], IDRiD [46], PAPILA [37], and Retina [30] in $\mathcal{D}_1$; NIH [54], CXP [26], ZhangCXR [34], RSNA [50], and SIIM-ARC [31] in $\mathcal{D}_2$; OCTID [16] in $\mathcal{D}_3$; ISIC2016 [18], ISIC2017 [8], and ISIC2018 [7, 52] in $\mathcal{D}_4$; NCH [33] in $\mathcal{D}_5$. These datasets span various tasks, including multi-class, binary-class, and multi-label disease classification as well as segmentation.

4.1.2. Training Setting.

In CoSMIC, at the t-th continual SSL pre-training stage, both $f_{\mathcal{D}_t}$ and $f_{\mathcal{D}_{t-1}}$ are initialized with the parameters learned from the $(t-1)$ stage. Notably, $f_{\mathcal{D}_{t-1}}$ is frozen, and no data from previous domains are required. The projection head, $h_{\mathcal{D}_t}$, and the decoder, $\mathbf{R}$, are trainable and are

used exclusively for $f_{\mathcal{D}_t}$. The default continual pre-training domain order is $\mathcal{D}_1 \rightarrow \mathcal{D}_2 \rightarrow \mathcal{D}_3 \rightarrow \mathcal{D}_4$.

During the pre-training phase, we adopt the AdamW optimizer with a weight decay set to 0.04 under the cosine scheduler, consistent with the training settings in DINO [2]. The learning rate undergoes a linear increase over the initial 6 epochs, attaining its base value based on the linear scaling rule: $lr = 1e{-}5 \times \frac{B}{256}$. The pre-training process for each domain is limited to a maximum of 100 epochs, conducted in a distributed manner with a batch size of 64 across 4 NVIDIA A6000 GPUs. Early stopping is implemented if the model's loss remains unchanged for 10 consecutive epochs. All other hyper-parameters and configurations align with those specified in DINO.

During the fine-tuning phase, we implement downstream tasks for multi-class and binary-class classification under RETFound settings [71], multi-label classification under PCRLv2 settings [70], and segmentation based on MMseg (version 0.25.0) [9], an open source toolbox based on PyTorch. More details are provided in the Suppl. 8.

4.1.3. Evaluation Metrics.

We use the following metrics to evaluate CoSMIC:

(i) **Performance**: We use the area under the ROC curve (AUC) and accuracy (ACC) for multi-class and binary classification, AUC for multi-label classification, and the Dice similarity coefficient (Dice) for segmentation tasks. To assess the transferability of models on downstream tasks in the domain $\mathcal{D}_t$, we calculate the average performance across all tasks within a single domain as: $A_{\mathcal{D}_t} = \frac{1}{\|\mathcal{D}_t\|}\sum_{i=1}^{\|\mathcal{D}_t\|} A^i_{\mathcal{D}_t}$, where $\|\mathcal{D}_t\|$ is the number of downstream tasks in $\mathcal{D}_t$, and $A^i_{\mathcal{D}_t}$ denotes the performance of the i-th tasks in $\mathcal{D}_t$. For each downstream task, we perform three independent runs and report the mean results along with the standard deviation. Statistical significance is assessed using an independent two-sample t-test.

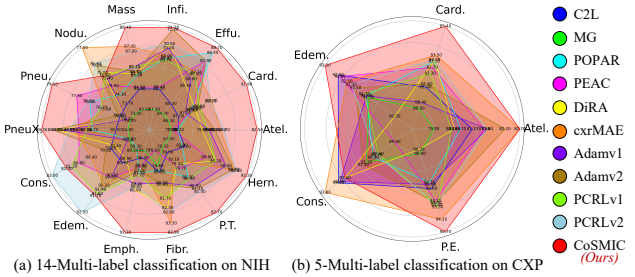
(ii) **Knowledge Forward** measures how much domain-specific knowledge the model has learned in $\mathcal{D}_t$. Mathematically: $KF_{\mathcal{D}_t} = A_{\mathcal{D}_t} - A_{\mathcal{D}_0}$, where $A_{\mathcal{D}_0}$ denotes the average performance of the ImageNet-pretrained ViT-B.

(iii) **Knowledge Backward** quantifies how much information the model has forgotten from the previous domain, namely: $KB_{\mathcal{D}_t} = \frac{1}{T-1}\sum_{i=1,i\neq t}^{T}(A_{\mathcal{D}_{t-1},i} - A_{\mathcal{D}_t,i})$. $A_{\mathcal{D}_t,i}$ is the performance of the model on domain i after pre-trained on domain t.

4.2. Comparison with State-of-the-arts

4.2.1. Comparing to Domain-specific Models

To evaluate the capability of domain-specific knowledge acquisition in CoSMIC, we fine-tune it on downstream tasks after each pre-training stage on $\mathcal{D}_t$. The results are compared to SOTA specific pre-training models in the respective domains. Results are presented in Table 2. Specifically, (i) in $\mathcal{D}_1$, we compare CoSMIC against 7 SSL models, including DiRA, Adamv2, RETFound, KeepFIT, FLAIR, ViLReF_ViT, and Uni4Eye++. Among these models, although CoSMIC exhibits slightly lower ACC or AUC performance on the Glaucoma, PAPILA, and Retina datasets compared to RETFound (pre-trained on 1M private fundus images) and KeepFIT (pre-trained on 2.3M fundus images), it still demonstrates competitive results using only 0.19M images for pre-training. Furthermore, compared to Flair, which is trained on 0.29M images, CoSMIC shows a significant performance improvement ($P<0.05$) across all datasets. (ii) In $\mathcal{D}_2$, CoSMIC exhibits exceptional performance, notably outperforming 10 SSL models, including C2L, MG, POPAR, PEAC, DiRA, cxrMAE, Adamv1/v2, and PCRLv1/v2, even when trained with limited annotations (using random selections of 1%, 10%, and 100% of the training data). The performance of each disease in multi-label classification on NIH and CXP is shown in Fig. 4. (iii) In $\mathcal{D}_3$, CoSMIC surpasses 6 models, including 2 OCT-specific models (LLCT and LACNN), and 4 OCT-SSL models (RETFound, Uni4Eye++, OCT-CoDA, and MIM_OCT). In this domain, CoSMIC surprisingly achieves higher performance compared to RETFound, which is pre-trained on 0.74M OCT images. (iv) In $\mathcal{D}_4$, CoSMIC continues its impressive performance, outperforming 3 skin-specific classification models: FTN, SSGNet, and SkinDistViT; and 3 skin-specific segmentation models: Attention-Unet, U-Net++, and Trans U-Net.



(a) 14-Multi-label classification on NIH (b) 5-Multi-label classification on CXP

Figure 4. **Disease-level comparison** of AUC scores (↑, %) with 100% labeling for multi-label classification on NIH and CXP.

Overall, compared to the aforementioned domain-specific pre-trained models, **CoSMIC achieves a powerful feature extraction capability, enabling it to acquire robust domain-specific knowledge within each medical field** under the continual SSL pre-training paradigm.

4.2.2. Comparing to Unified Pre-training Models

To evaluate the generalization capability of the proposed CoSMIC, we assess its performance on 15 downstream tasks across 4 domains after completing the entire continual pre-training from $\mathcal{D}_1$ to $\mathcal{D}_4$. Specifically, we compare CoSMIC with (i) the performance when training from *Scratch*; (ii) *Joint* pre-training $\{\mathcal{D}_t\}_{t=1}^4$ using $\mathcal{L}_{CoSMIC}$; (iii) base-

Methods	Retinoscopy ($\mathcal{D}_1$)									
	APTOS		Glaucoma		IDRiD		PAPILA		Retina	
	ACC(%,↑)	AUC(%,↑)	ACC(%,↑)	AUC(%,↑)	ACC(%,↑)	AUC(%,↑)	ACC(%,↑)	AUC(%,↑)	ACC(%,↑)	AUC(%,↑)
DiRA† [20]	90.97±0.11	91.82±0.04	87.65±0.03	92.09±0.02	81.50±0.01	79.17±0.07	86.17±0.22	76.17±0.10	76.14±0.04	79.17±0.12
Adamv2† [51]	87.27±0.06	90.43±0.10	86.13±0.22	90.16±0.11	79.62±0.02	77.68±0.04	85.69±0.29	74.62±0.24	78.17±0.01	77.17±0.02
RETFound [71]	92.34±0.05	94.48±0.12	**90.22±0.11**	94.34±0.13	82.11±0.14	83.01±0.06	87.23±0.04	84.51±0.14	86.37±0.11	85.24±0.21
KeepFIT [56]	91.34±0.03	93.73±0.10	87.42±0.05	93.05±0.01	85.05±0.13	81.53±0.15	86.70±0.19	**88.22±0.07**	84.90±0.10	**88.80±0.02**
FLAIR [48]	92.18±0.01	93.42±0.01	88.34±0.11	93.92±0.30	82.11±0.25	83.00±0.19	87.48±0.08	83.24±0.08	85.79±0.04	85.74±0.01
ViLReF_ViT [59]	86.39±0.03	79.80±0.03	81.94±0.13	82.91±0.06	80.49±0.01	81.84±0.13	84.57±0.01	71.10±0.17	84.71±0.01	82.00±0.14
Uni4Eye++ [1]	87.55±0.07	86.94±0.02	83.44±0.05	90.15±0.09	78.68±0.09	64.79±0.06	84.57±0.01	74.55±0.02	83.06±0.05	78.92±0.08
CoSMIC(ours)	**93.00±0.12**	**95.39±0.05**	89.46±0.25	**94.38±0.11**	**85.84±0.22**	**83.97±0.23**	**88.11±0.07**	80.10±0.27	**87.54±0.06**	87.28±0.10

Methods	Radiography($\mathcal{D}_2$)																	
	NIH			CXP			ZhangCXR			RSNA			SIIM-ARC					
	1% AUC(%,↑)	10% AUC(%,↑)	100% AUC(%,↑)	1% AUC(%,↑)	10% AUC(%,↑)	100% AUC(%,↑)	1% AUC(%,↑)	10% AUC(%,↑)	100% AUC(%,↑)	1% AUC(%,↑)	10% AUC(%,↑)	100% AUC(%,↑)	1% Dice(%,↑)	10% Dice(%,↑)	100% Dice(%,↑)			
C2L [68]	59.69±0.04	73.09±0.03	81.45±0.08	80.86±0.03	82.50±0.08	88.20±0.10	86.25±0.03	90.77±0.15	95.24±0.08	79.88±0.01	84.97±0.09	89.88±0.31	49.09±0.30	63.93±0.02	75.85±0.41			
MG [74]	59.19±0.17	70.09±0.11	80.75±0.13	79.20±0.02	82.14±0.03	86.70±0.08	85.92±0.15	89.76±0.08	96.88±0.04	80.43±0.15	85.34±0.14	90.87±0.10	49.88±0.66	62.51±0.20	73.49±0.76			
POPAR [44]	61.11±0.02	72.06±0.13	82.25±0.03	80.58±0.01	82.98±0.03	87.72±0.12	88.75±0.34	93.47±0.03	97.08±0.07	81.88±0.10	87.02±0.10	91.36±0.14	51.39±0.77	68.04±0.64	78.08±0.53			
PEAC [75]	60.96±0.01	72.64±0.05	82.41±0.01	80.44±0.05	83.38±0.02	87.96±0.05	89.42±0.02	93.55±0.17	96.45±0.20	81.69±0.01	86.69±0.10	91.43±0.02	50.75±0.40	65.90±0.56	77.41±0.14			
DiRA† [20]	62.62±0.04	75.14±0.08	82.79±0.05	81.50±0.09	83.76±0.02	87.46±0.02	90.54±0.17	94.16±0.04	96.78±0.07	82.19±0.02	87.24±0.02	91.88±0.17	49.26±0.70	67.49±0.75	80.73±0.18			
cxrMAE [57]	65.38±0.02	76.06±0.01	84.40±0.02	82.50±0.10	85.40±0.29	89.32±0.02	90.97±0.05	95.62±0.06	97.92±0.22	84.97±0.01	88.27±0.12	92.74±0.01	52.17±0.08	68.86±0.20	83.70±0.21			
Adamv1† [25]	63.64±0.03	74.40±0.01	82.79±0.15	81.74±0.14	84.10±0.13	87.78±0.05	88.27±0.12	92.87±0.11	96.42±0.11	83.14±0.14	85.71±0.09	92.34±0.15	44.34±0.84	59.72±1.68	72.53±0.60			
Adamv2† [51]	64.63±0.12	75.28±0.05	83.23±0.24	82.42±0.08	84.88±0.02	88.14±0.06	90.36±0.22	94.25±0.04	97.55±0.06	83.36±0.13	87.85±0.18	91.88±0.13	48.48±0.56	64.82±0.42	75.26±0.30			
PCRLv1 [69]	64.18±0.03	75.79±0.07	83.05±0.05	80.90±0.09	83.50±0.10	87.70±0.07	87.94±0.08	93.39±0.09	97.04±0.04	82.67±0.18	85.68±0.04	90.27±0.03	50.27±0.71	65.82±0.48	75.32±0.37			
PCRLv2 [70]	65.02±0.10	77.19±0.13	83.95±0.09	82.08±0.15	84.12±0.17	88.24±0.22	89.77±0.18	94.39±0.14	97.86±0.04	84.24±0.03	87.24±0.17	91.32±0.24	51.36±0.09	67.66±0.02	77.29±0.30			
CoSMIC(ours)	**69.81±0.09**	**80.73±0.09**	**85.11±0.05**	**83.80±0.04**	**86.50±0.21**	**89.78±0.03**	**91.58±0.01**	**96.41±0.20**	**99.81±0.15**	**86.41±0.17**	**90.67±0.11**	**93.10±0.21**	**55.03±0.58**	**69.37±1.37**	**86.90±0.89**			

Methods	Ophthalmoscope($\mathcal{D}_3$)		Methods	Dermoscope($\mathcal{D}_4$)						
	OCTID			ISIC2016		ISIC2017		ISIC2018		ISIC2018(seg)
	ACC(%,↑)	AUC(%,↑)		ACC(%,↑)	AUC(%,↑)	ACC(%,↑)	AUC(%,↑)	ACC(%,↑)	AUC(%,↑)	Dice(%,↑)
LLCT‡ [55]	92.32±0.21	96.47±0.11	FTN‡ [23]	90.77±0.21	93.89±0.02	92.67±0.04	93.30±0.19	91.53±0.17	92.47±0.09	-
LACNN‡ [13]	94.97±0.29	95.14±0.02	SSGNet‡ [11]	90.50±0.17	92.72±0.04	91.50±0.02	92.74±0.13	87.90±0.12	91.59±0.03	-
RETFound [71]	95.94±0.21	98.34±0.11	SkinDistViT‡ [39]	91.29±0.21	94.99±0.27	93.33±0.20	96.57±0.22	92.59±0.27	93.47±0.07	-
Uni4Eye++ [1]	93.16±0.16	97.32±0.15	Attention-Unet‡ [42]	-	-	-	-	-	-	87.59±0.33
OCT-CoDA [53]	95.14±0.38	97.97±0.04	U-Net++ ‡ [73]	-	-	-	-	-	-	87.86±0.17
MIM_OCT [45]	94.23±0.15	96.41±0.08	Trans U-Net‡ [4]	-	-	-	-	-	-	88.06±0.22
CoSMIC(ours)	**96.68±0.18**	**99.31±0.05**	CoSMIC(ours)	**93.67±0.20**	**96.50±0.20**	**94.33±0.22**	**97.61±0.04**	**93.45±0.04**	**95.92±0.20**	**88.24±0.13**

Table 2. **Comparison with domain-specific models.** The best performance is **bold** and the second best is underlined. †: The pre-trained weights are reimplemented using their official code. ‡: The model is not built on SSL pre-training but is specifically designed as a domain-specific model. All other performance results are based on our reimplementation using publicly available pre-trained weights.

line: ImageNet-supervised ViT-B [12]; (iv) general SSL baselines: DINO [2] and MAE [22]; (v) continual SSL: CaSSLe [14] and MedCoss [63]; and (vi) medical foundation models: LVM-Med [41], RadImageNet [40], UniMed-CLIP [35], and Unimiss+ [58]. For the models in (iii), (iv), and (vi), we reimplement them using their publicly available pre-trained parameters. For the models in (v), we pre-train them using their official code, adhering to CoSMIC's default continual pre-training domain order.

The comparison results are shown in Fig. 5, where CoSMIC significantly outperforms these competitors with a notable improvement (P<0.05), demonstrating that **CoSMIC exhibits exceptional generalization capability for multi-domain medical image analysis**.

4.2.3. Comparing to Continual SSL Models

To evaluate the transferability and historical knowledge retention capabilities of our CoSMIC, we analyze its effectiveness using the $A_{\mathcal{D}_t}$, $KF_{\mathcal{D}_t}$, and $KB_{\mathcal{D}_t}$ metrics. We compare these results with the continual SSL methods: CaSSLe [14] and MedCoss [63]. All three methods begin with $\mathcal{D}_0$, whose parameters are pre-trained on ImageNet. Then they are pre-trained from $\mathcal{D}_1$ to $\mathcal{D}_4$ sequentially. We also evaluate these models on an unseen domain, Histopathology, to examine their generalization on future domains. As illustrated in Fig. 6, the results show that, compared to CaSSLe (distillation-based), CoSMIC achieves higher KF scores, indicating its superior domain-specific knowledge acquisition. It also retains knowledge as effectively as MedCoss (rehearsal-based). However, when incrementing from $\mathcal{D}_0$ to $\mathcal{D}_1$, where data form $\mathcal{D}_0$ is unavailable, MedCoss suffers significant performance degradation ($KB_{\mathcal{D}_1} = 7.33$), while CoSMIC avoid this with $KB_{\mathcal{D}_1} = 0.21$. On the unseen Histopathology domain ($\mathcal{D}_5$), CoSMIC shows excellent generalization, with only a 0.71 performance drop compared to $\mathcal{D}_0$, demonstrating its **strong transferability to future domains** by effectively mitigating catastrophic forgetting.

More visualizations and detailed statistical results are provided in Suppl. 9 and Suppl. 10.

4.3. Ablation Studies

We conduct ablation studies to assess the effectiveness of different components in CoSMIC. Results are in Table 3, with **row 4** showing the **default setting**. (i) Removing $\mathcal{L}_{Recon}$ **(row 1)** leads to a decline in the KF metric, reflecting the effectiveness of $\mathcal{L}_{Recon}$ to acquire domain-specific knowledge. (ii) Omitting the similarity calculation for either $([XZ]_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$ **(row 2)** or $(X_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$ **(row 3)** in $\mathcal{L}_{CoCst}$ causes a significant loss in knowledge reten-

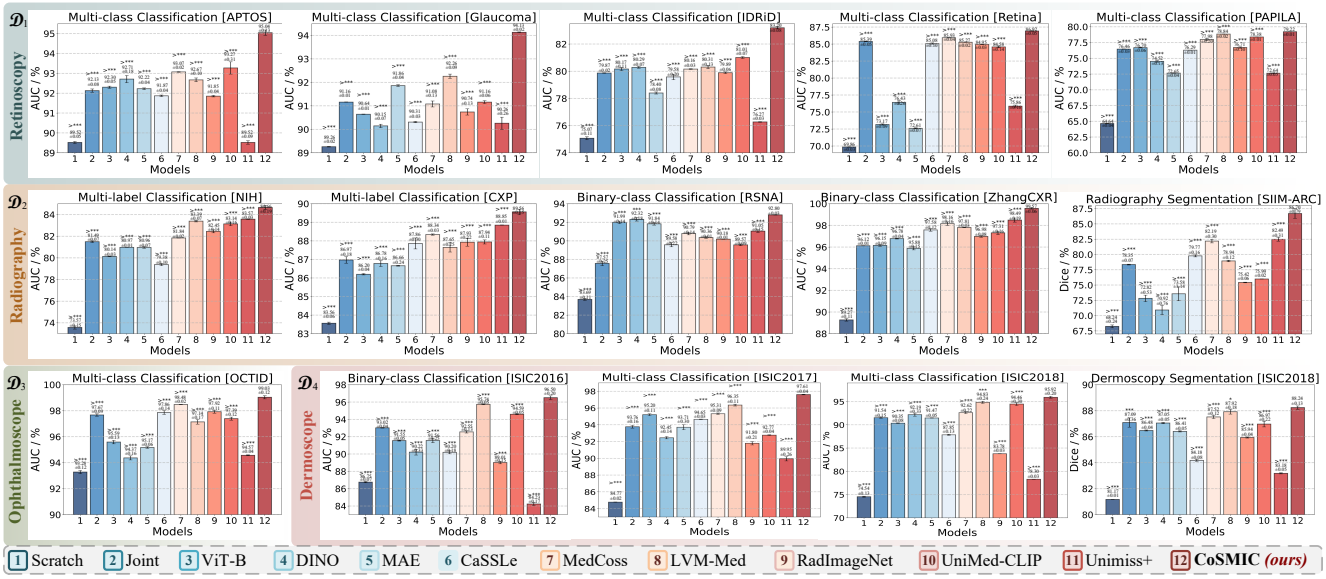


Figure 5. **Comparison with unified pre-training models.** P-values, calculated using a two-sided t-test, are displayed in the figure, where 'n.s.': No Significance, '*': p<0.05, '**': p<0.01, '***': p<0.001, and '>***': p<0.0001.

#	$\mathcal{L}_{Recon}$	$\mathcal{L}_{CoCst}$ $([XZ]_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$	$(X_{\mathcal{D}_t}, Y_{\mathcal{D}_t})$	$\mathcal{D}_1$ $A(\uparrow)$	$KF(\uparrow)$	$KB(\downarrow)$	$\mathcal{D}_2$ $A(\uparrow)$	$KF(\uparrow)$	$KB(\downarrow)$	$\mathcal{D}_3$ $A(\uparrow)$	$KF(\uparrow)$	$KB(\downarrow)$	$\mathcal{D}_4$ $A(\uparrow)$	$KF(\uparrow)$	$KB(\downarrow)$	$\mathcal{D}_5$ $A(\uparrow)$	$KF(\uparrow)$
1	✗	✓	✓	85.79	3.77	0.21	88.30	3.31	0.12	97.91	2.64	0.18	93.15	2.25	0.13	94.06	-0.66
2	✓	✗	✓	78.64	4.07	8.07	76.91	3.46	2.36	90.14	3.08	2.16	93.03	2.13	2.61	85.29	-9.43
3	✓	✓	✗	83.63	3.32	1.12	86.47	3.08	0.78	96.49	2.38	0.74	92.17	1.27	0.87	91.92	-2.80
4	✓	✓	✓	87.70	5.62	0.21	90.66	5.53	0.16	99.03	3.72	0.15	94.57	3.67	0.14	94.01	-0.71
5	w/o AGC			85.04	4.37	0.91	88.91	4.53	0.57	98.32	3.34	0.51	93.96	3.06	0.56	92.84	-1.88
6	w/o Hierarchical Alignment			86.59	4.85	0.44	89.42	4.55	0.29	98.68	3.31	0.34	93.49	2.59	0.27	93.00	-1.72
7	$\mathcal{D}_4 \rightarrow \mathcal{D}_3 \rightarrow \mathcal{D}_2 \rightarrow \mathcal{D}_1$			87.32	4.72	0.24	90.11	4.84	0.18	99.00	3.75	0.13	94.71	4.37	0.13	94.03	-0.69

Table 3. **Ablation studies of proposed CoSMIC**. We assess component efficacy on $A(\uparrow)$, $KF(\uparrow)$, and $KB(\downarrow)$.

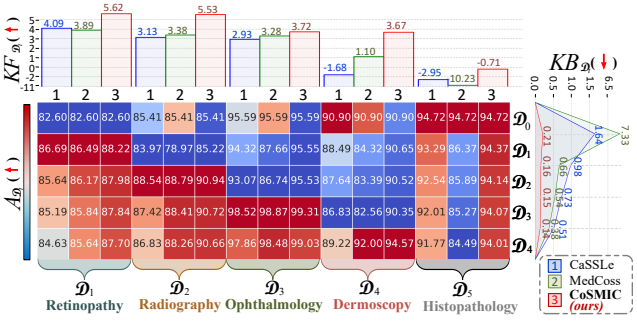


Figure 6. **Comparison with continual SSL models.**

tion. While the model attempts to learn new knowledge in each domain, catastrophic forgetting severely hinders its overall performance. (iii) When eliminating **AGC (row 5)** and predicting $[XZ]_{\mathcal{D}_t}$ by directly adding $X_{\mathcal{D}_t}$ and $Z_{\mathcal{D}_t}$, CoSMIC's ability to retain old knowledge declines, as the noise in $Z_{\mathcal{D}_t}$ interferes with the optimization of $f_{\mathcal{D}_t}$. (iv) Without **Hierarchical Alignment (row 6)**, CoSMIC's performance noticeably reduced, indicating its importance in learning multi-scale features. (v) Experimenting with a **Different Pre-training Order (row 7)** shows only a slight performance drop (less than 0.4 A), suggesting that CoSMIC's ability to mitigate catastrophic forgetting is not influenced by the pre-training order.

5. Conclusion

We propose a medical foundation model called CoSMIC. It employs a continual self-supervised learning paradigm to enhance both data-specific and generalized representations for multi-domain medical image analysis. The core idea of CoSMIC is the proposed $\mathcal{L}_{CoCst}$, which maximizes the conditional mutual information between different views of the model's current representation based on the knowledge learned from previous domains. This approach enables CoSMIC to effectively balance the acquisition of domain-specific features with the preservation of previously learned knowledge. Extensive experiments demonstrate the effectiveness of CoSMIC. It achieves the SOTA performance on 15 downstream tasks across 4 medical domains, outperforming 29 domain-specific models and 11 unified models, including four foundational medical models. This robust performance underscores CoSMIC's exceptional capability in applications across multiple healthcare domains.

6. Acknowledgments

Supported in part by the National Natural Science Foundation of China under Grant 62171323, 62271155, in part by the Yeqisun Joint Funds of the National Natural Science Foundation of China under Grant U2441252, in part by National Key R&D Program of China under Grant 2020YFA0711400, in part by Shanghai Municipal Science and Technology Major Project (2021SHZDZX0100), in part by the Changjiang Scholars Program of China, in part by the Fundamental Research Funds for the Central Universities.

References

[1] Zhiyuan Cai, Li Lin, Huaqing He, Pujin Cheng, and Xiaoying Tang. Uni4eye++: A general masked image modeling multi-modal pre-training framework for ophthalmic image classification and segmentation. *IEEE Transactions on Medical Imaging*, 2024. 2, 7

[2] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9650–9660, 2021. 2, 3, 5, 6, 7

[3] Hyuntak Cha, Jaeho Lee, and Jinwoo Shin. Co2l: Contrastive continual learning. In *Proceedings of the IEEE/CVF International conference on computer vision*, pages 9516–9525, 2021. 2, 4

[4] Jieneng Chen, Yongyi Lu, Qihang Yu, Xiangde Luo, Ehsan Adeli, Yan Wang, Le Lu, Alan L Yuille, and Yuyin Zhou. Transunet: Transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306*, 2021. 7

[5] Ziyu Chen, Hanli Wang, and Chang Wen Chen. Self-supervised video representation learning by serial restoration with elastic complexity. *IEEE Transactions on Multimedia*, 26:2235–2248, 2023. 1, 3

[6] Haoyang Cheng, Haitao Wen, Xiaoliang Zhang, Heqian Qiu, Lanxiao Wang, and Hongliang Li. Contrastive continuity on augmentation stability rehearsal for continual self-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5707–5717, 2023. 2

[7] Noel Codella, Veronica Rotemberg, Philipp Tschandl, M Emre Celebi, Stephen Dusza, David Gutman, Brian Helba, Aadi Kalloo, Konstantinos Liopyris, Michael Marchetti, et al. Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic). *arXiv preprint arXiv:1902.03368*, 2019. 5

[8] Noel CF Codella, David Gutman, M Emre Celebi, Brian Helba, Michael A Marchetti, Stephen W Dusza, Aadi Kalloo, Konstantinos Liopyris, Nabin Mishra, Harald Kittler, et al. Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic). In *2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)*, pages 168–172, 2018. 5

[9] MMSegmentation Contributors. MMSegmentation: Openmmlab semantic segmentation toolbox and benchmark. https://github.com/open-mmlab/mmsegmentation, 2020. 6

[10] Coen De Vente, Koenraad A Vermeer, Nicolas Jaccard, He Wang, Hongyi Sun, Firas Khader, Daniel Truhn, Temirgali Aimyshev, Yerkebulan Zhanibekuly, Tien-Dung Le, et al. Airogs: Artificial intelligence for robust glaucoma screening challenge. *IEEE Transactions on Medical Imaging*, 43 (1):542–557, 2023. 5

[11] Baoping Dong, Xu Fu, and Xiufeng Kang. Ssgnet: semi-supervised multi-path grid network for diagnosing melanoma. *Pattern Analysis and Applications*, 26(1):357–366, 2023. 7

[12] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, and Sylvain and Gelly. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. 3, 5, 7

[13] Leyuan Fang, Chong Wang, Shutao Li, Hossein Rabbani, Xiangdong Chen, and Zhimin Liu. Attention to lesion: Lesion-aware convolutional neural network for retinal optical coherence tomography image classification. *IEEE transactions on medical imaging*, 38(8):1959–1970, 2019. 7

[14] Enrico Fini, Victor G Turrisi Da Costa, Xavier Alameda-Pineda, Elisa Ricci, Karteek Alahari, and Julien Mairal. Self-supervised models are continual learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9621–9630, 2022. 2, 4, 7

[15] Qiankun Gao, Chen Zhao, Yifan Sun, Teng Xi, Gang Zhang, Bernard Ghanem, and Jian Zhang. A unified continual learning framework with general parameter-efficient tuning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11483–11493, 2023. 2

[16] Peyman Gholami, Mohana Kuppuswamy Parthasarathy, Priyanka Roy, and Vasudevan Lakshminarayanan. Octid citation. https://doi.org/10.5683/SP2/W43PFI, 2018. 5

[17] Alex Gomez-Villa, Bartlomiej Twardowski, Lu Yu, Andrew D Bagdanov, and Joost Van de Weijer. Continually learning self-supervised representations with projected functional regularization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3867–3877, 2022. 2, 4

[18] David Gutman, Noel CF Codella, Emre Celebi, Brian Helba, Michael Marchetti, Nabin Mishra, and Allan Halpern. Skin lesion analysis toward melanoma detection: A challenge at the international symposium on biomedical imaging (isbi) 2016, hosted by the international skin imaging collaboration (isic). *arXiv preprint arXiv:1605.01397*, 2016. 5

[19] Fatemeh Haghighi, Mohammad Reza Hosseinzadeh Taher, Michael B Gotway, and Jianming Liang. Dira: Discriminative, restorative, and adversarial learning for self-supervised medical image analysis. In *Proceedings of the IEEE/CVF*

Conference on Computer Vision and Pattern Recognition, pages 20824–20834, 2022. 1

[20] Fatemeh Haghighi, Mohammad Reza Hosseinzadeh Taher, Michael B Gotway, and Jianming Liang. Self-supervised learning for medical image analysis: Discriminative, restorative, or adversarial? *Medical Image Analysis*, 94:103086, 2024. 2, 7

[21] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9729–9738, 2020. 2, 4

[22] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022. 2, 3, 7

[23] Xinzi He, Ee-Leng Tan, Hanwen Bi, Xuzhe Zhang, Shijie Zhao, and Baiying Lei. Fully transformer network for skin lesion analysis. *Medical Image Analysis*, 77:102357, 2022. 7

[24] Carlos Hernández-Pérez, Marc Combalia, Sebastian Podlipnik, Noel CF Codella, Veronica Rotemberg, Allan C Halpern, Ofer Reiter, Cristina Carrera, Alicia Barreiro, Brian Helba, et al. Bcn20000: Dermoscopic lesions in the wild. *Scientific Data*, 11(1):641, 2024. 5

[25] Mohammad Reza Hosseinzadeh Taher, Michael B Gotway, and Jianming Liang. Towards foundation models learned from anatomy in medical imaging via self-supervision. In *MICCAI Workshop on Domain Adaptation and Representation Transfer*, pages 94–104, 2023. 1, 7

[26] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghgoo, Robyn Ball, Katie Shpanskaya, et al. Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI conference on artificial intelligence*, pages 590–597, 2019. 5

[27] Alistair EW Johnson, Tom J Pollard, Nathaniel R Greenbaum, Matthew P Lungren, Chih-ying Deng, Yifan Peng, Zhiyong Lu, Roger G Mark, Seth J Berkowitz, and Steven Horng. Mimic-cxr-jpg, a large publicly available database of labeled chest radiographs. *arXiv preprint arXiv:1901.07042*, 2019. 5

[28] Dahuin Jung, Dongyoon Han, Jihwan Bang, and Hwanjun Song. Generating instance-level prompts for rehearsal-free continual learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11847–11857, 2023. 2

[29] Kaggle. Diabetic retinopathy detection. https://www.kaggle.com/c/diabetic-retinopathy-detection, . 5

[30] Kaggle. Retina dataset. https://www.kaggle.com/datasets/jr2ngb/cataractdataset/data, . 5

[31] Kaggle. Siim-acr pneumothorax segmentation. https://www.kaggle.com/c/siim-acr-pneumothorax-segmentation, . 5

[32] Karthik, Maggie, and Sohier Dane. Aptos 2019 blindness detection. https://kaggle.com/competitions/aptos2019-blindness-detection, 2019. Kaggle. 5

[33] Jakob Nikolas Kather, Niels Halama, and Alexander Marx. 100,000 histological images of human colorectal cancer and healthy tissue. https://doi.org/10.5281/zenodo.1214456, 2018. 5

[34] Daniel S. Kermany, Michael Goldbaum, Wenjia Cai, et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5):1122–1131.e9, 2018. 5

[35] Muhammad Uzair Khattak, Shahina Kunhimon, Muzammal Naseer, Salman Khan, and Fahad Shahbaz Khan. Unimed-clip: Towards a unified image-text pretraining paradigm for diverse medical imaging modalities. *arXiv preprint arXiv:2412.10372*, 2024. 1, 2, 7

[36] Ungsoo Kim. Machine learn for glaucoma. https://doi.org/10.7910/DVN/1YRRAC, 2018. Harvard Dataverse. 2, 5

[37] Oleksandr Kovalyk, Juan Morales-Sánchez, Rafael Verdú-Monedero, Inmaculada Sellés-Navarro, Ana Palazón-Cabanes, and José-Luis Sancho-Gómez. Papila. https://doi.org/10.6084/m9.figshare.14798004.v1, 2022. Dataset Available at Figshare. 5

[38] Yihang Liu, Lianghua He, Ying Wen, Longzhen Yang, and Hongzhou Chen. Afire: Anatomy-driven self-supervised learning for fine-grained representation in radiographic images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 18996–19005, 2025. 1

[39] Vlad-Constantin Lungu-Stan, Dumitru-Clementin Cercel, and Florin Pop. Skindistilvit: Lightweight vision transformer for skin lesion classification. In *International Conference on Artificial Neural Networks*, pages 268–280. Springer, 2023. 7

[40] Xueyan Mei, Zelong Liu, Philip M. Robson, Brett Marinelli, Mingqian Huang, Amish Doshi, Adam Jacobi, Chendi Cao, Katherine E. Link, Thomas Yang, Ying Wang, Hayit Greenspan, Timothy Deyer, Zahi A. Fayad, and Yang Yang. Radimagenet: An open radiologic deep learning research dataset for effective transfer learning. *Radiology: Artificial Intelligence*, 0:e210315, 0. 1, 2, 7

[41] Duy MH Nguyen, Hoang Nguyen, Nghiem Diep, Tan Ngoc Pham, Tri Cao, Binh Nguyen, Paul Swoboda, Nhat Ho, Shadi Albarqouni, Pengtao Xie, et al. Lvm-med: Learning large-scale self-supervised vision models for medical imaging via second-order graph matching. *Advances in Neural Information Processing Systems*, 36, 2024. 1, 2, 7

[42] Ozan Oktay. Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*, 2018. 7

[43] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. 4, 5

[44] Jiaxuan Pang, Fatemeh Haghighi, DongAo Ma, Nahid Ul Islam, Mohammad Reza Hosseinzadeh Taher, Michael B Gotway, and Jianming Liang. Popar: Patch order prediction and appearance recovery for self-supervised medical image analysis. In *MICCAI Workshop on Domain Adaptation and Representation Transfer*, pages 77–87, 2022. 7

[45] Theodoros Pissas, Pablo Márquez-Neila, Sebastian Wolf, Martin Zinkernagel, and Raphael Sznitman. Masked image modelling for retinal oct understanding. In *International Workshop on Ophthalmic Medical Image Analysis*, pages 115–125. Springer, 2024. 1, 2, 7

[46] Prasanna Porwal, Samiksha Pachade, Ravi Kamble, Manesh Kokare, Girish Deshmukh, Vivek Sahasrabuddhe, and Fabrice Meriaudeau. Indian diabetic retinopathy image dataset (idrid). https://dx.doi.org/10.21227/H25W98, 2018. 5

[47] Veronica Rotemberg, Nicholas Kurtansky, Brigid Betz-Stablein, Liam Caffery, Emmanouil Chousakos, Noel Codella, Marc Combalia, Stephen Dusza, Pascale Guitera, David Gutman, et al. A patient-centric dataset of images and metadata for identifying melanomas using clinical context. *Scientific Data*, 8(1):34, 2021. 5

[48] Julio Silva-Rodriguez, Hadi Chakor, Riadh Kobbi, Jose Dolz, and Ismail Ben Ayed. A foundation language-image model of the retina (flair): Encoding expert knowledge in text supervision. *Medical Image Analysis*, 99:103357, 2025. 1, 2, 7

[49] James Seale Smith, Leonid Karlinsky, Vyshnavi Gutta, Paola Cascante-Bonilla, Donghyun Kim, Assaf Arbelle, Rameswar Panda, Rogerio Feris, and Zsolt Kira. Coda-prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11909–11919, 2023. 2

[50] Anouk Stein, MD, Carol Wu, Chris Carr, George Shih, Jamie Dulkowski, kalpathy, Leon Chen, Luciano Prevedello, Marc Kohli, MD, Mark McDonald, Peter, Phil Culliton, Safwan Halabi MD, and Tian Xia. Rsna pneumonia detection challenge, 2018. 5

[51] Mohammad Reza Hosseinzadeh Taher, Michael B Gotway, and Jianming Liang. Representing part-whole hierarchies in foundation models by learning localizability composability and decomposability from anatomy via self supervision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11269–11281, 2024. 2, 7

[52] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5(1):1–9, 2018. 5

[53] Lehan Wang, Chongchong Qi, Chubin Ou, Lin An, Mei Jin, Xiangbin Kong, and Xiaomeng Li. Multieye: Dataset and benchmark for oct-enhanced retinal disease recognition from fundus images. *IEEE Transactions on Medical Imaging*, 2024. 2, 7

[54] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M Summers. Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2097–2106, 2017. 5

[55] Huajie Wen, Jian Zhao, Shaohua Xiang, Lin Lin, Chengjian Liu, Tao Wang, Lin An, Lixin Liang, and Bingding Huang. Towards more efficient ophthalmic disease classification and lesion location via convolution transformer. *Computer Methods and Programs in Biomedicine*, 220:106832, 2022. 7

[56] Ruiqi Wu, Chenran Zhang, Jianle Zhang, Yi Zhou, Tao Zhou, and Huazhu Fu. Mm-retinal: Knowledge-enhanced foundational pretraining with fundus image-text expertise. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 722–732, 2024. 1, 2, 7

[57] Junfei Xiao, Yutong Bai, Alan Yuille, and Zongwei Zhou. Delving into masked autoencoders for multi-label thorax disease classification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3588–3600, 2023. 1, 2, 3, 7

[58] Yutong Xie, Jianpeng Zhang, Yong Xia, and Qi Wu. Unimiss+: Universal medical self-supervised learning from cross-dimensional unpaired data. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):10021–10035, 2024. 1, 2, 7

[59] Shengzhu Yang, Jiawei Du, Jia Guo, Weihang Zhang, Hanruo Liu, Huiqi Li, and Ningli Wang. Vilref: An expert knowledge enabled vision-language retinal foundation model. *arXiv preprint arXiv:2408.10894*, 2024. 2, 7

[60] Xu Yang, Xuan Chen, Moqi Li, Kun Wei, and Cheng Deng. A versatile framework for continual test-time domain adaptation: Balancing discriminability and generalizability. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23731–23740, 2024. 2, 4

[61] Fei Ye and Adrian G Bors. Online task-free continual generative and discriminative learning via dynamic cluster memory. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26202–26212, 2024. 2

[62] Jin Ye, Guoan Wang, Yanjun Li, Zhongying Deng, Wei Li, Tianbin Li, Haodong Duan, Ziyan Huang, Yanzhou Su, Benyou Wang, et al. Gmai-mmbench: A comprehensive multimodal evaluation benchmark towards general medical ai. *Advances in Neural Information Processing Systems*, 37: 94327–94427, 2025. 2

[63] Yiwen Ye, Yutong Xie, Jianpeng Zhang, Ziyang Chen, Qi Wu, and Yong Xia. Continual self-supervised learning: Towards universal multi-modal medical data representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11114–11124, 2024. 1, 2, 3, 7

[64] Jiazuo Yu, Yunzhi Zhuge, Lu Zhang, Ping Hu, Dong Wang, Huchuan Lu, and You He. Boosting continual learning of vision-language models via mixture-of-experts adapters. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23219–23230, 2024. 2

[65] Mengyao Zhai, Lei Chen, Frederick Tung, Jiawei He, Megha Nawhal, and Greg Mori. Lifelong gan: Continual learning for conditional image generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2759–2768, 2019. 2

[66] Jie Zhang, Chen Chen, Weiming Zhuang, and Lingjuan Lyu. Target: Federated class-continual learning via exemplar-free

distillation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4782–4793, 2023. 2

[67] Hongbo Zhao, Bolin Ni, Junsong Fan, Yuxi Wang, Yuntao Chen, Gaofeng Meng, and Zhaoxiang Zhang. Continual forgetting for pre-trained vision models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28631–28642, 2024. 2

[68] Hong-Yu Zhou, Shuang Yu, Cheng Bian, Yifan Hu, Kai Ma, and Yefeng Zheng. Comparing to learn: Surpassing imagenet pretraining on radiographs by comparing image representations. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part I 23*, pages 398–407, 2020. 1, 2, 4, 7

[69] Hong-Yu Zhou, Chixiang Lu, Sibei Yang, Xiaoguang Han, and Yizhou Yu. Preservational learning improves self-supervised medical image models by reconstructing diverse contexts. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3499–3509, 2021. 1, 2, 7

[70] Hong-Yu Zhou, Chixiang Lu, Chaoqi Chen, Sibei Yang, and Yizhou Yu. A unified visual information preservation framework for self-supervised pre-training in medical image analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(7):8020–8035, 2023. 1, 3, 6, 7, 2

[71] Yukun Zhou, Mark A Chia, Siegfried K Wagner, Murat S Ayhan, Dominic J Williamson, Robbert R Struyven, Timing Liu, Moucheng Xu, Mateo G Lozano, Peter Woodward-Court, et al. A foundation model for generalizable disease detection from retinal images. *Nature*, 622(7981):156–163, 2023. 1, 2, 3, 6, 7

[72] Yuhang Zhou, Haolin Li, Siyuan Du, Jiangchao Yao, Ya Zhang, and Yanfeng Wang. Low-rank knowledge decomposition for medical foundation models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11611–11620, 2024. 1

[73] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. Unet++: A nested u-net architecture for medical image segmentation. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop, DLMIA 2018, and 8th International Workshop, ML-CDS 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 20, 2018, Proceedings 4*, pages 3–11. Springer, 2018. 7

[74] Zongwei Zhou, Vatsal Sodha, Jiaxuan Pang, Michael B Gotway, and Jianming Liang. Models genesis. *Medical Image Analysis*, 67:101840, 2021. 2, 3, 7

[75] Ziyu Zhou, Haozhe Luo, Jiaxuan Pang, Xiaowei Ding, Michael Gotway, and Jianming Liang. Learning anatomically consistent embedding for chest radiography. In *BMVC: proceedings of the British Machine Vision Conference. British Machine Vision Conference*, 2023. 7